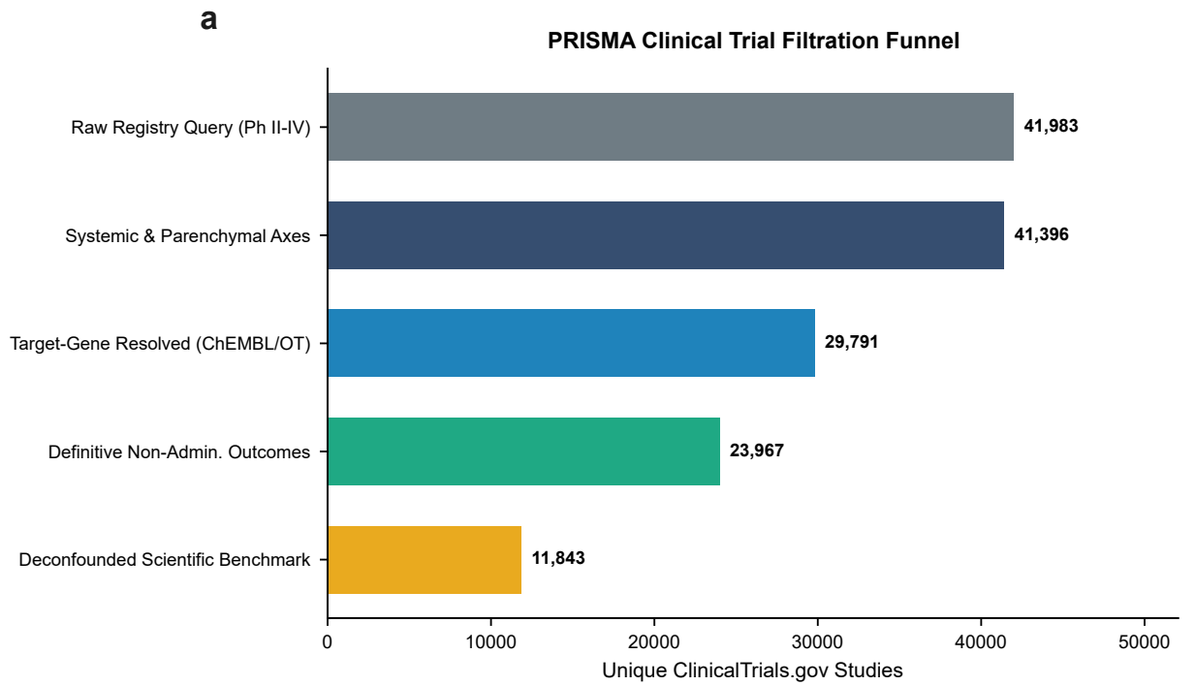
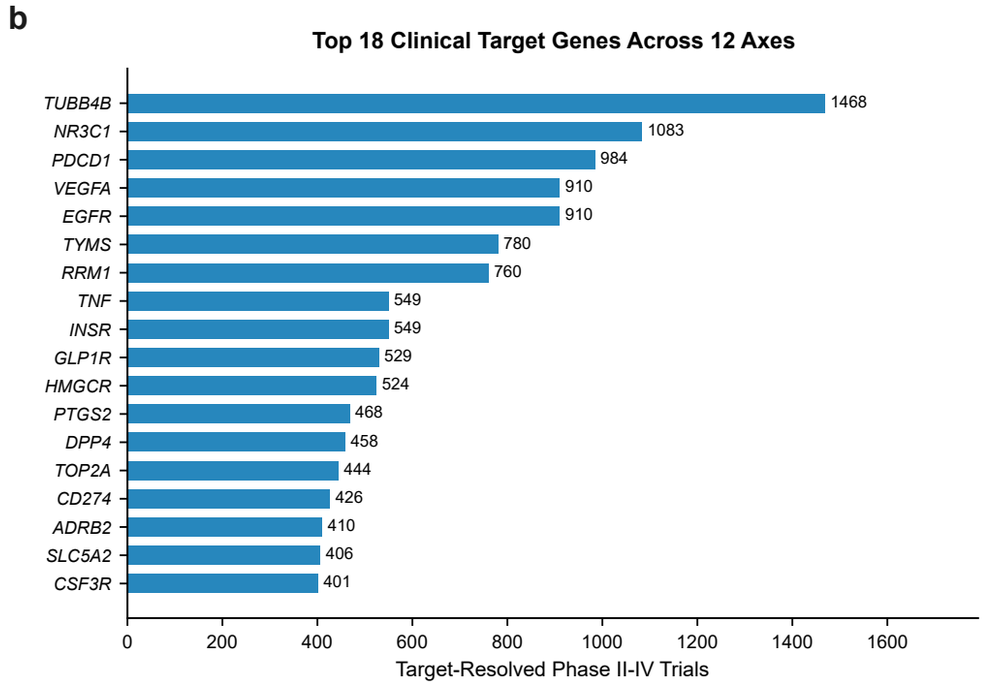
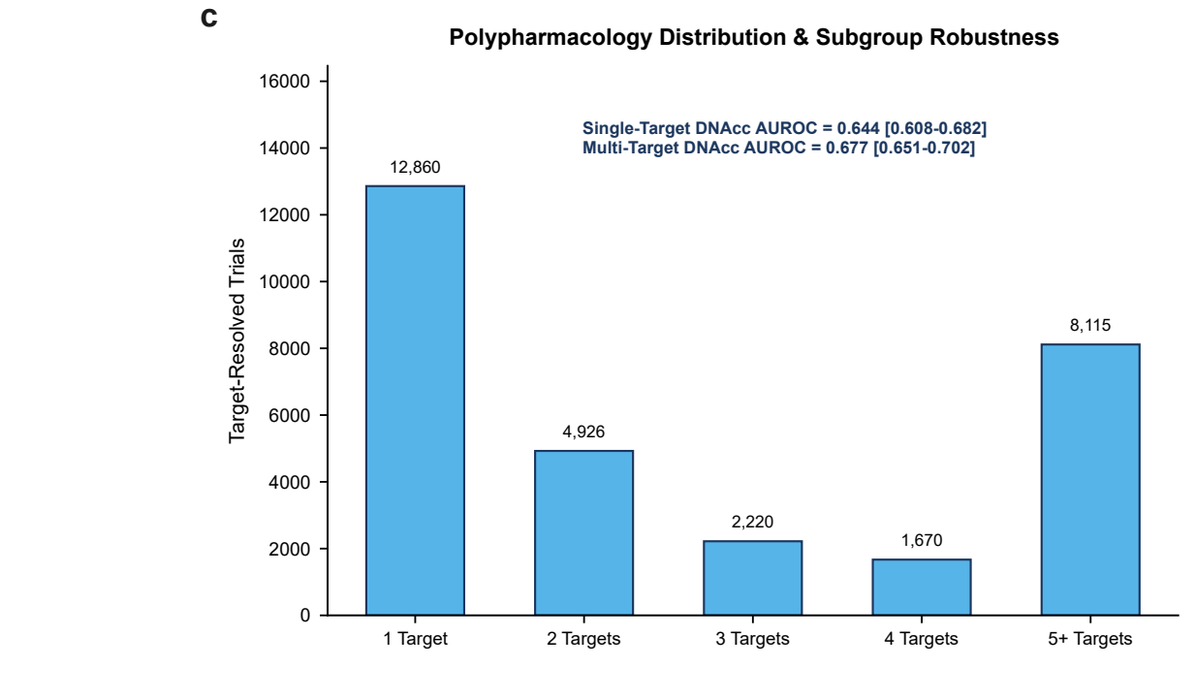
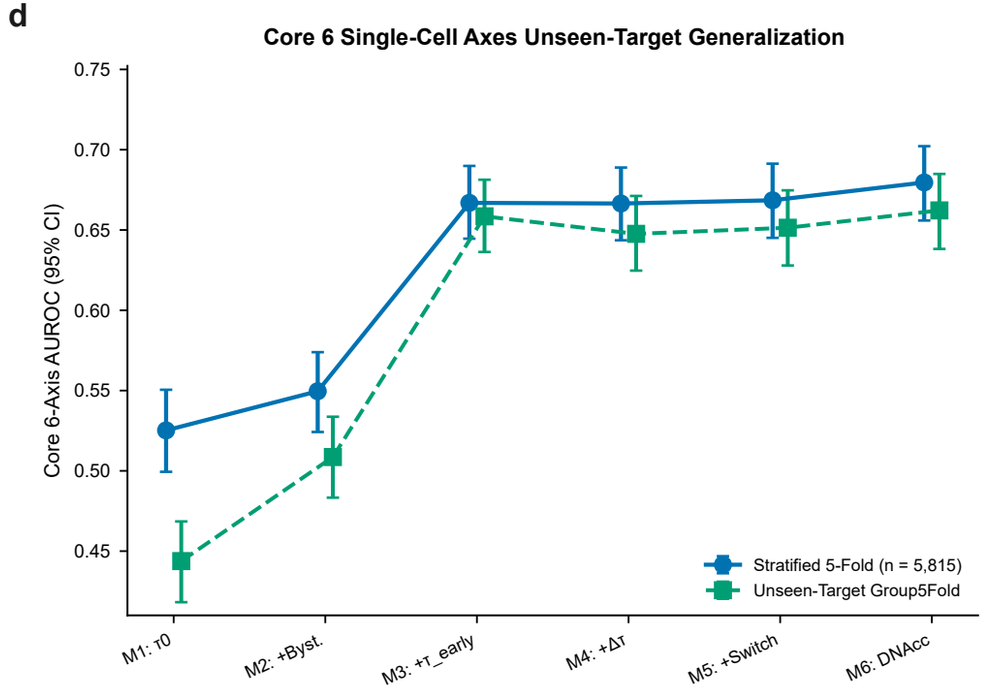

a
PRISMA Clinical Trial Filtration Funnel
Raw Registry Query (Ph II-IV)
41,983
Systemic & Parenchymal Axes
41,396
Target-Gene Resolved (ChEMBL/OT)
29,791
Definitive Non-Admin. Outcomes
23,967
Deconfounded Scientific Benchmark
11,843
0
10000
20000
30000
40000
50000
Unique ClinicalTrials.gov Studies


b
Top 18 Clinical Target Genes Across 12 Axes
TUBB4B 1468
NR3C1 1083
PDCD1 984
VEGFA 910
EGFR 910
TYMS 780
RRM1 760
TNF 549
INSR 549
GLP1R 529
HMGCR 524
PTGS2 468
DPP4 458
TOP2A 444
CD274 426
ADRB2 410
SLC5A2 406
CSF3R 401
0
200
400
600
800
1000
1200
1400
1600
Target-Resolved Phase II-IV Trials


c
Polypharmacology Distribution & Subgroup Robustness
Single-Target DNAcc AUROC = 0.644 [0.608-0.682]
Multi-Target DNAcc AUROC = 0.677 [0.651-0.702]
Target-Resolved Trials
16000
14000
12000
10000
8000
6000
4000
2000
0
12,860
4,926
2,220
1,670
8,115
1 Target
2 Targets
3 Targets
4 Targets
5+ Targets


d
Core 6 Single-Cell Axes Unseen-Target Generalization
Core 6-Axis AUROC (95% CI)
0.75
0.70
0.65
0.60
0.55
0.50
0.45
Stratified 5-Fold (n = 5,815)
Unseen-Target Group5Fold
M1: τ0
M2: +Byst.
M3: +τ_early
M4: +Δτ
M5: +Switch
M6: DNAcc